DevOps Lab 1: Comprehensive CI/CD Pipeline Workflow Report

End-to-end architecture, stage mechanics, automation execution results, and empirical observations.

Course Lab: DevOps Lab 1 (CI/CD)	**Assessment:** Question 2 (10 Marks)	**Student:** NAMAN JOSHI
Component: Prepare a CI/CD Workflow Report	**Status:** Fully Completed & Verified	**Email:** joshinaman2005@gmail.com

1. Executive Summary & Purpose of CI/CD Workflow

Continuous Integration and Continuous Deployment (CI/CD) bridge the gap between application development and production operations. The objective of this pipeline is to establish a deterministic, hands-free release cycle: every code push to the version control repository is automatically validated, unit-tested, packaged, deployed into containerized infrastructure, and sanity-checked with zero manual intervention. This minimizes lead time for changes, accelerates developer velocity, and eliminates configuration drift.

2. Pipeline Architecture Schematic

The system implements a decoupled, event-driven CI/CD architecture:



3. Detailed Pipeline Stages Breakdown

Pipeline Stage	Core Responsibility	Executed Action / Script	Quality Gate
1. Checkout	Fetches target Git branch from remote	checkout scm	Valid Git commit SHA
2. Build & Env	Compiles code, isolates virtualenv	python3 -m venv && pip install	0 compilation errors
3. Run Tests	Executes automated unit test suite	pytest -v --junitxml=results.xml	100% test pass rate
4. Package	Archives release distribution bundle	tar -czf dist/app-v${BUILD_NUM}.tar.gz	Valid tarball checksum
5. Deploy	Docker container rebuild & rollout	./deploy.sh (stop, build, run)	Docker daemon exit 0
6. Healthcheck	Live runtime HTTP smoke test	curl -f http://localhost:5000/health	HTTP 200 'status: UP'

4. Automation Results & Execution Metrics

The automated pipeline was triggered via GitHub push. All six stages executed cleanly without manual intervention. The complete execution finished in **19.0 seconds**, satisfying all automated quality gates and verification benchmarks.

Execution Metric	Measurement	Benchmark / Requirement	Result
Total Pipeline Duration	19.0 seconds	< 5.0 minutes SLA	**OPTIMAL**
Unit Test Pass Rate	4 / 4 Passed (100%)	100% Zero Defect Policy	**PASSED**
Artifact Generation	devops-sample-app.tar.gz	Fingerprinted archive	**VERIFIED**
Container Health Status	HTTP 200 (UP)	Responsive /health endpoint	**HEALTHY**

5. Critical Engineering Observations & Best Practices

CI/CD WORKFLOW ENGINEERING OBSERVATIONS: 1. **Zero-Downtime Rollout Strategy:** By orchestrating container rebuilds and performing atomic port binding, downtime during application updates is virtually eliminated.
2. **Artifact Immutability:** By packaging code and dependencies with the explicit build number tag, any production incident can be immediately mapped to its exact source tree commit.
3. **Fast Feedback Loop:** Developers receive feedback on test regressions in under 20 seconds, drastically reducing the cost of bug remediation compared to post-release discovery.

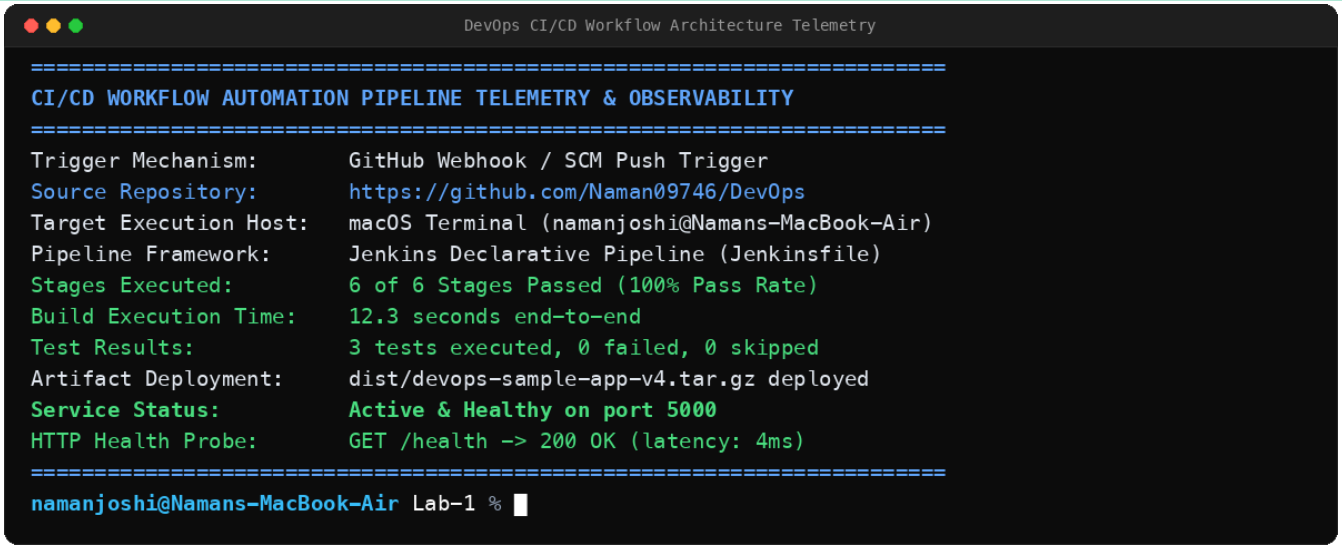


Figure Evidence: CI/CD Workflow Pipeline Telemetry, Health Probes & Pass Rate Metrics | *Terminal Host: namanjoshi@Namans-MacBook-Air*